

Econ 131
Fall 2020

Final Exam

December 2020

Student Name:

Student ID:

Exam Instructions

- **Explanation should be written using pens (we recommend black or blue ink, as these often scan the best).** No pencils, except for graphs.
- **Show your work.** Credit will only be awarded on the basis of what is written on the exam.
- **Affirm the academic honesty pledge below.** For those writing on a non-printed copy, please just write "Academic Honesty Pledge as on exam", and sign your name.
If you do not affirm this pledge, your exam will be marked invalid.

0. ACADEMIC HONESTY PLEDGE

I confirm that I have abided by all academic honesty rules for UC Berkeley and Economics 131. I confirm that I did not see this exam before my official exam start time. I confirm that I have not shared and will not share this exam with anyone else.

Signature: _____

1. True/False/Uncertain Questions (20 points total)

Explain your answer fully based on what was discussed in class, since all the credit is based on the explanation. Your grade depends entirely on the substance of your justification, not on whether you are correct in writing “True”, “False” or “Uncertain”. Note that it is possible to answer each question for full credit with three sentences or fewer, and answers longer than ten lines long will not be graded.

- (a) Suppose that a friend calls me, waking me from a nap. I then look out the window and see that my neighbor is drowning in their pool and about to die. I run out of the house and rescue the neighbor.

True/False: My friend caused the neighbor to survive just as much as I did. *(2 points)*

- (b) **True/False/Uncertain:** A business income tax can raise tax revenue without reducing investment. *(2 points)*

- (c) **True/False/Uncertain:** A capital tax on savings must reduce savings. *(2 points)*

(d) **True/False/Uncertain:** When there is uncertainty in the marginal cost of pollution reduction and when the marginal benefit to reducing pollution declines quickly in the amount of pollution reduced (i.e., when the social marginal benefit curve for pollution reduction is steep, such as for reducing mercury content in tap water), tradable permits generate less deadweight loss than corrective tax. *(2 points)*

(e) In class, we watched a video of Supreme Court Justice Antonin Scalia asserting that government provision of benefits to the poor reduces private charitable provision of benefits to the poor.

True/False/Uncertain: There is no evidence to support his assertion. *(2 points)*

(f) **True/False/Uncertain:** There is evidence suggesting “individual failures” and/or a lack of information in higher education choices. *(2 points)*

(g) **True/False/Uncertain:** If having car insurance (i.e., insurance that provides cash payments in the event of an expensive car accident) makes individuals drive more recklessly, that is an example of moral hazard. *(2 points)*

(h) **True/False/Uncertain:** The Affordable Care Act (i.e., “ObamaCare”) can be justified on redistributive grounds rather than efficiency (i.e., market-failure-correcting) grounds. *(2 points)*

(i) **True/False/Uncertain:** Disability insurance causes some non-employment. *(2 points)*

(j) **True/False/Uncertain:** An unfunded pension system like Social Security is worse than a funded pension system. *(2 points)*

2. Optimal Savings Rates (5 points)

3. Externalities (9 points)

In 2021, the hoverboards promised to us in *Back to the Future II* are finally invented. Experts know that using them will cause some amount of pollution, and they turn to you to decide how to correct for that externality. Suppose that the demand for hoverboards is given by

$$P = 250 - \frac{1}{4}Q_D$$

$$P = Q_S$$

- (a) Suppose that most experts assume that each hoverboard consumed will cause an externality equal to \$100 of pollution damage over its lifetime of use.
- (i) Suggest a quantity cap that could be imposed to reach the socially optimum number of hoverboards sold at equilibrium (assume that no tax is used). State the quantity specifically. *(2 points)*
- (ii) Suggest a tax that could be implemented to reach the socially optimum number of hoverboards sold at equilibrium. Be sure to specify on whom the tax is levied and the amount of the tax. *(1 point)*
- (iii) Would the tax you suggest cause any inefficiency in the market? Explain. *(1 point)*
- (iv) Assume the government sold a limited number of permits for hoverboard use equal to the quantity suggested in part (i) at auction. How much would the permits sell for? *(1 point)*

- (b) As new data on hoverboard use comes in, scientists realize that hoverboards create less pollution than they expected, creating only \$50 of pollution damage over a lifetime of use. How much deadweight loss would be created if the government had imposed the tax suggested in part (a)(ii)? (*Note*: Even if you were unable to work out the prior parts of this question, you can provide an answer using assumed quantities for partial credit.) (4 points)

4. Disability Insurance (20 points)

Let's suppose that each individual in the economy earns \$100. There is no public provision of disability insurance, however, and so they receive \$0 when sick. Let's start out assuming that there are four types of people, each with a different utility function and lifestyle-based disability probability q_i (where i indexes the different riskiness groups l, m, h, o):

- Type L: $q_l = 0.2$, $u(c) = \sqrt{c}$
- Type M: $q_m = 0.75$, $u(c) = \sqrt{c}$
- Type H: $q_h = 0.8$, $u(c) = c$
- Type O: $q_o = 0.5$, $u(c) = c^2$

Let's suppose each individual can purchase disability insurance from private firms at premium price p , providing a lump-sum benefit b if they end up disabled. For (a) and (b), we'll assume that there is no asymmetric information; i.e. insurance companies know each consumer's type, and can offer specialized plan options to each type. There are 50 people of each type in the economy.

- (a) Briefly state **and justify** whether each type would benefit from, be indifferent to, or worse off from receiving insurance at actuarially fair premiums (compared to receiving no insurance). (2 points)

For the rest of the problem, assume that types H and O have migrated out of the economy, so only types L and M remain.

- (b) Maintaining our assumption of no asymmetric information, let's assume that private providers only offer actuarially fair plan policies with full insurance. Calculate the benefit b and price p offered to each type (note again that types H and O have migrated out of the economy, so only types L and M remain). (2 points)

For the rest of the problem, let's relax the "no asymmetric information" assumption: instead, suppose that insurance providers cannot observe worker types.

- (c) Assume that there is only one insurance company (so it may earn a profit) and it offers only one insurance plan. It offers a plan with a benefit of $b = 100$ for a price of $p = 75$. Call this insurance plan Plan #1. Which types will choose to buy it? *Hint: calculating utilities rather than maximum-prices-willing-to-pay will help you save time later. (4 points)*
- (d) Will the insurance company still be able to offer Plan #1 in a perfectly competitive equilibrium? (*Note: Even if you were unable to work out the prior parts of this question, you can provide an intuitive answer for partial credit using general results from section/lecture.*) (1 point)
- (e) A new firm decides to offer a second plan so that there are now two plans offered in the economy: Plan #1 and the new plan called Plan #2. Plan #2 has benefit $b = 10$. What is the price for Plan #2 if the new firm expects both types to buy it and the new firm sets an actuarially fair price? (1 point)

- (f) At this price for Plan #2, which types will buy Plan #2, if any? Which types will buy Plan #1, if any? Which will buy no plan at all, if any? (*Hint 1:* You've computed several of the necessary utilities for this problem in (c)). (*Note:* Even if you were unable to work out the prior parts of this question, you can provide an intuitive answer for partial credit using general results from section/lecture.). (*4 points*)

- (g) Entrepreneurs learn that the new firm was making a profit under Plan #2. Those entrepreneurs start lots of new firms so that any firm offering a plan with benefit $b = 10$ has to set the premium equal to actual average costs. As a result, Plan #2 disappears, but there is a new plan called Plan #3 with $b = 10$ and premium p equal to the plan's actual average costs. Assume that the types who used to buy Plan #2 (computed in the previous problem) buy Plan #3 and are the only ones who buy it. What premium will Plan #3 have? (*1 point*)

(h) Suppose that only Plan #1 and Plan #3 are offered in the economy. Which types (if any) will buy Plan #1? Which types (if any) will buy Plan #3? Which types (if any) will buy no plan? (*Note: Even if you were unable to work out the prior parts of this question, you can provide an intuitive answer for partial credit using results from section/lecture.*) (*3 points*)

(i) Let social welfare equal the sum of all individuals' expected utility. Under which economic scenario is social welfare maximized: the economic scenario of part (b) or the economic scenario of (h)? (*Note: You may but definitely do not have to use algebra in your answer. Even if you were unable to work out the prior parts of this question, you can provide an intuitive answer for partial credit using results from section/lecture results.*) (*2 points*)

Reminder: please sign the Academic Honesty Pledge on the first page!